HBC Diabetes Diary

v11.1 Type 1 Diabetes APP

User manual and installation guide

For people living with insulin-treated diabetes, their relatives and their treating physicians.

July 2026

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1. Introduction: what this application is and who it is for

The HBC Diabetes Diary is a personal health diary for people with insulin-treated diabetes. It runs on phones, tablets, laptops and desktop computers, works without an internet connection, and all data remains on the user's own device. The diary records blood glucose readings, the carbohydrate content of meals and the insulin administered, then turns them into clear charts and statistics.

The application provides five key services that a paper diary cannot. First: it performs calculations for the user — for example, it suggests a pre-meal insulin dose based on values agreed with the treating physician. Second: it gives immediate feedback — for example, it issues a warning when a low blood glucose value is recorded. Third: it can be shared, so a designated follower — a relative or the treating physician — can view fresh readings live on their own phone. Fourth: it is bilingual (Hungarian and English), and the blood glucose unit can be switched as well. Fifth — new in v9.0 —: it produces a printable medical report for any freely chosen date range, which can be taken to an appointment as a PDF.

What the application does not do

The application is not a medical device and does not replace medical advice. All of its calculations — including the insulin dose suggestion and the estimated HbA1c and GMI values — are for information only. Dose decisions must in every case follow the guidance of the treating physician. The application displays this warning in the appropriate places as well.

Important medical warning

In case of feeling unwell, first measure blood glucose and act according to the learned protocol; attend to logging only afterwards. Example: at a reading of 3.2 mmol/l, first take 15 grams of fast-acting carbohydrate (3 glucose tablets or 150 ml of fruit juice), re-check after 15 minutes, and record the value only after that.

2. What is new in v11.1

This chapter briefly summarises the changes compared to v10.0. Every feature is described in detail in the later chapters; diary data remains unchanged after the update. The features of earlier versions (v10.0, v9.0 and v8.0) are summarised below.

- **Editing foods after saving:** the foods belonging to an already recorded meal can now be added to or removed afterwards in the entry's edit window (pencil icon); the carbohydrate total is recalculated automatically.
- **More precise sync error messages:** when the Sync now button is triggered manually, the application now reports the exact cause of a failure (expired login, missing permission, deleted file, no internet connection, invalid Client ID); the scheduled, automatic sync keeps retrying silently in the background.
- **App name on the home screen:** the installed app icon now shows the label "HBC Diab Napló" underneath.
- **Public Google access:** Google authorisation for Drive sync is now open to anyone, without a test-user list (see chapter 7.2 for details).

Earlier version (v10.0) features (in brief)

v10.0 introduced the following new features compared to v9.0; these remain available in v11.1.

- **Larger desktop button bar:** the top page-selector buttons now share the content width of the app evenly (not the full monitor width), with larger text and icons — easier to tap, and they resize proportionally when the window is resized.
- **Improved hamburger menu:** the mobile menu panel now always opens BELOW the header and never covers it; the panel scrolls when it contains many items.
- **Custom background color:** next to Settings → Appearance, a background color can be chosen from a pop-up color palette; it appears as a soft gradient throughout the application — in both light and dark mode — while card contrast is preserved; it can be cleared at any time with the Reset button.

Earlier version (v9.0) features (in brief)

v9.0 introduced the following new features compared to v8.0; these remain available in v10.0 and v11.1.

- **Restored program files:** several files of the v8.0 package (app.js, i18n.js) were saved in a damaged, truncated state, so the application started incorrectly or not at all. v9.0 contains complete, verified code.
- **Mobile hamburger menu:** the ☰button in the top right corner of the header makes every page reachable (Overview, Charts, Entries, Statistics, Foods, Sync, Settings, New entry). The quick bar with the five most-used pages remains at the bottom of the screen — so the patient, the family member and the physician can all navigate easily, for example during a demonstration or when reviewing statistics.
- **Desktop top bar:** every page-selector and settings button is visible on every page in the bar under the header — in the well-proven way known from v7.

- **Larger, adaptive logo:** the TypeOneDiab logo is 20% larger in desktop view and scales automatically with the window width; on mobile its maximum size adapts to the space available in the header.
- **Insulins from drop-down lists:** in Settings the rapid (bolus) insulins available in Hungary (Humalog, NovoRapid, Fiasp, Apidra, Lyumjev, Actrapid…) and the basal insulins (Lantus, Abasaglar, Toujeo, Levemir, Tresiba, Semglee…) can be chosen from a list; a custom name can still be entered.
- **Custom color picker:** in addition to the Light / Dark / Automatic appearance, the accent color of the application can now be chosen freely from a pop-up color palette (Settings → Color theme → Custom color).
- **Alerts only for same-day values:** the family-member alert fires only for the latest measurement taken on the current day; back-dated, older values never trigger an alert. The shared data package also includes the CGM readings, so the alert always reflects the current state.
- **Language completeness:** every label on every page appears in both Hungarian and English, according to the selected language.

Earlier v8.0 features (in brief)

v9.0 contains every v8.0 feature. The main v8.0 additions compared to v7.0 were the following; diary data remains untouched after every update.

- Refreshed appearance: the Type 1 Diabetes ribbon logo appears in the header and in the application icons, and the emoji icons have been replaced by a consistent, drawn (SVG) icon set.
- Bottom navigation bar on phones: the tabs are located in a thumb-friendly bar at the bottom of the screen. On desktop computers the familiar top tab row remains.
- Medical report (PDF): from the Statistics tab, a one-page printable summary is generated for any freely chosen from–to period — TIR, estimated HbA1c, GMI, variability measures, charts, daily pattern and the detailed log.
- GMI in Statistics: alongside the estimated HbA1c, the internationally used Glucose Management Indicator is displayed as well.
- Daily pattern chart: hourly average, minimum and maximum values for the selected period — it clearly reveals, for example, early-morning rises.
- Configurable time-of-day boundaries: the hour limits for the default meal type (Breakfast…Late-night snack) and for the bolus calculator's time periods can be adjusted to the user's daily routine in Settings.
- Typo protection: before saving an unusual blood glucose value (below 2.0 mmol/l or above 25.0 mmol/l), the application requests confirmation.
- Larger touch targets, better dark-mode contrast and keyboard focus indication — more comfortable and more accessible operation on every device.
- Bug fixes: a rare crash of the weekly TIR tile has been eliminated, and entry cards now display the insulin names configured in Settings.

3. Glossary: the terms used in this manual

The table explains every term that appears in this manual or in the application. When encountering an unfamiliar word, its meaning can be looked up here quickly.

Term	Plain-language meaning
Blood glucose (BG)	The sugar level of the blood. In Hungary it is measured in mmol/l, e.g. 5.6 mmol/l. In the American unit, mg/dl, the same value is 101 mg/dl.
mmol/l and mg/dl	Two units for the same value. Conversion: 1 mmol/l = 18 mg/dl. It can be switched on the Settings tab.
Carbs (CH)	The blood-glucose-raising component of food, measured in grams. Example: a bread roll is about 30 grams of carbs.
Bolus	The rapid-acting insulin taken with a meal, e.g. Humalog. It covers the carbohydrate content of the food.
Basal	The once-daily long-acting insulin, e.g. Lantus. It works steadily in the background all day.
ICR (carb-to-insulin ratio)	Indicates how many grams of carbohydrate 1 unit of rapid insulin covers. Example: with an ICR of 10, a lunch containing 60 grams of carbs requires 6 units of insulin. The value may differ by time of day.
Sensitivity (ISF)	Indicates how much 1 unit of rapid insulin lowers blood glucose. Example: with a sensitivity of 2.5 mmol/l, a reading of 9.0 mmol/l is expected to drop to 6.5 mmol/l after 1 unit.
IOB (insulin on board)	The amount of previously injected insulin still active. Example: 2 hours after injecting 4 units, about 2.3 units are still working. The application subtracts this from the next suggestion, protecting against stacked doses.
TIR (time in range)	The percentage of readings that fell within the target band (by default between 3.9 and 10.0 mmol/l). Example: 78% TIR means almost eight out of ten readings were in range. Medical guidance generally considers above 70% good.
HbA1c	The proportion of sugar bound to red blood cells, reflecting the average blood glucose of the past 2–3 months. The application computes an estimate from the readings; the precise value comes only from a laboratory test.

Term	Plain-language meaning
GMI	Glucose Management Indicator: an HbA1c-like percentage computed from average glucose with an international formula. A common reference number in the CGM world — also an estimate, not a laboratory result.
SD and CV	Variability measures in the medical report. SD (standard deviation) shows how far readings scatter around the average; CV is the same as a percentage — below 36% is generally considered stable.
CGM	Continuous glucose monitor sensor (e.g. FreeStyle Libre or Dexcom) measuring every minute or every 5 minutes. The sensor's factory export file can be imported into the application.
PWA	Progressive web application: a website that can be installed on phones and computers like an app, with an icon, full screen and offline operation.
Google Drive	Google's free cloud storage (15 GB with every Google account). The application saves a copy of the diary there when sync is enabled.
Sync	Automatic reconciliation of data between two devices. Here: uploading the diary to Drive and downloading fresh values to the follower's phone.
Owner and follower	The two roles of sharing. The owner is the user keeping the diary; the follower is the person — a relative or the treating physician — who can view the shared diary live, in read-only mode.
Hypoglycaemia (hypo)	Blood glucose lower than normal — in the application, a value below the configured lower limit (3.9 mmol/l by default). Symptoms may include trembling, sweating, weakness. Action: 15 grams of fast carbs, re-check after 15 minutes.
Hyperglycaemia (hyper)	Blood glucose higher than normal — in the application, a value above the upper limit (10.0 mmol/l by default). The application marks it red on the charts and adds a correction to the bolus suggestion.
JSON and CSV	Two file formats. JSON is the complete, restorable backup of the diary. CSV is a table that Excel opens, for medical review or personal analysis.

4. The application view by view

On a phone, a quick bar at the bottom of the screen holds the five most-used pages (Overview, Charts, New entry, Entries, Statistics), while the ☰ hamburger menu in the top right corner of the header reaches every page — including Foods, Sync and Settings. On desktop computers and tablets all seven tabs are visible in a row under the header on every page, and the new-entry button sits in the top right corner of the header. This chapter introduces each view: its purpose, its contents, and its recommended use.

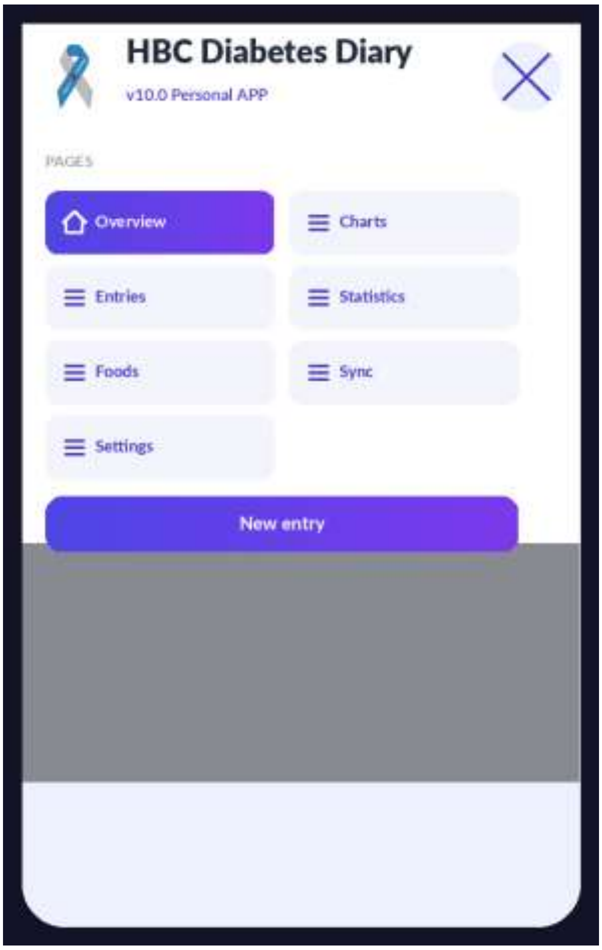
4.0 Screenshots: what the application looks like

The following screenshots show the key elements using the application's real colors, logo and labels, so that using it feels familiar from the very first moment.

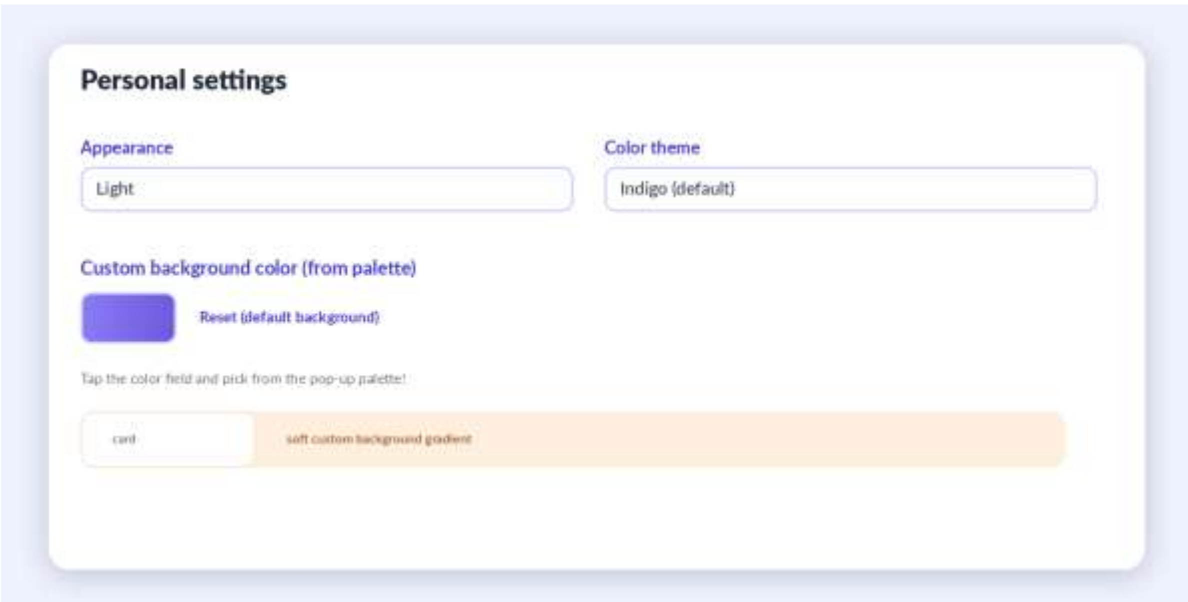
a) Desktop header + top button bar (v10 — larger, evenly split)

b) Mobile bottom bar + hamburger button

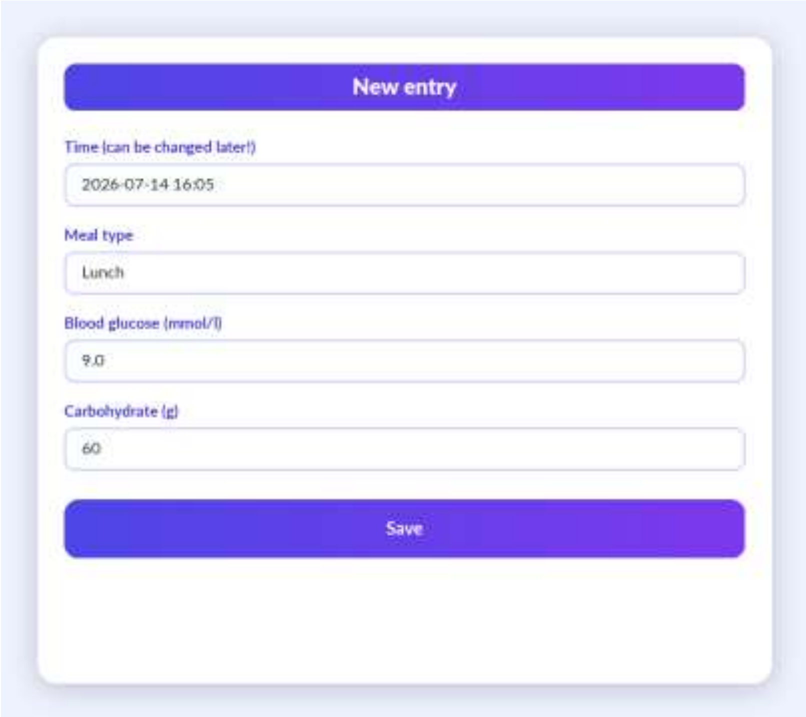
c) Opened hamburger menu (v10 — opens BELOW the header)



d) Appearance / Color theme setting — custom background color picker



e) New entry form (date / time field)



f) Entry cards with indicator colors (green / amber / red)

4.1 Overview (Dashboard)

This is the home screen, the summary of the day. At the top there are three motivation tiles: Streak counts how many days the diary has been kept complete (e.g. 12 days of complete logging), Today's logging circle shows how many of the 4 daily entries are done (e.g. 2/4), and the Weekly TIR goal shows time in range against the 70% target. Below them are three coloured summary tiles: insulin taken today, the 7-day glucose average and today's carbs. The largest card shows the latest blood glucose value: green if in range, amber if low, red if high. A short note for the physician can also be added here, for example: woke at 3 am, sweating. In the evening the application indicates if no basal insulin has been logged that day.

4.2 Charts

Three charts show the recent period: blood glucose (line chart), carbohydrate and insulin (bar charts). The range is switched with buttons: Day, Week, Month, or a Custom period where the start and end dates can be set freely (from–to). On the glucose curve a green band marks the target range, and the colour of each point grades the reading: green in range, amber low, red high. Imported CGM data appears on a separate chart. A practical example: before an appointment, set the range to Month and review the behaviour of the early-morning hours together with the physician.

4.3 Entries

The list of all records in chronological order, filtered by day or by any from–to period, and also by type (Meal, Check, Basal, Other activity). Every card shows the time, blood glucose, carbs, insulin and note. With the pencil icon any entry can be corrected later; the bin icon deletes it. Example: if it turns out in the evening that 54 grams were recorded at lunch instead of 45, it can be corrected here with two taps.

4.4 Statistics

A numeric summary for the chosen period, set with buttons (1, 7, 30, 90 days) or with a custom from–to date range: average glucose, TIR percentage, estimated HbA1c, GMI, and the distribution of readings (low, normal, high). A separate card shows the currently active insulin (IOB) and the actual carb-to-insulin ratio computed from the diary. Next to the estimated HbA1c and the GMI the application notes: this number is an estimate, not a laboratory result. An example of interpretation: a 90-day TIR of 82% with an estimated HbA1c of 6.8% indicates good balance, to be confirmed by the next laboratory test.

New in v9.0 is the Daily pattern chart: it groups the readings of the selected period by hour of day and shows the hourly average together with the band of the lowest and highest values. At a glance it reveals in which parts of the day blood glucose tends to rise or fall — for example the dawn phenomenon or a regularly under-estimated dinner.

Creating the medical report (PDF)

The Medical report card is located at the bottom of the Statistics tab. Set the start and end dates (from–to), then press Create report. A print-ready summary opens on a new page: the nickname (if configured) and the period in the header, followed by the key figures (number of readings, average, SD/CV, TIR, estimated HbA1c, GMI, low and high events, daily average insulin and carbs), the glucose curve of the whole period, the daily pattern, and finally the detailed log table. Use the Print button on the opened page to print on paper, or choose Save as PDF in the print dialog to create a file that can be sent by e-mail. The report is generated in the application's language and with the configured unit.

4.5 Foods

The user's own food database. It starts with 22 foods and their carb values, for example: a 1 cm slice of sourdough loaf is 15 grams of carbs, a medium apple is 15 grams. It can be extended at any time by entering a name, carb value and unit (e.g. cottage-cheese pancake, 25 grams, 1 pc). Foods added here appear in the quick picker of the New entry screen, so at mealtime they can be added with a single tap.

4.6 Sync

All data backup and sharing features are located here: setting up Google Drive sync (details in chapter 7), importing a CGM file (chapter 8), full backup to a JSON file, CSV download for Excel or for the physician, and text copy-paste between two devices. JSON is the complete, restorable copy of the diary; CSV is a tabular format that Excel opens. Recommendation: once a month press JSON Export and store the file — that is the backup. If the latest backup is older than 7 days, the Overview screen issues a warning.

4.7 Settings

The personal settings come in four groups. The first group is appearance: language (Hungarian or English), blood glucose unit (mmol/l or mg/dl), nickname, colour theme (indigo, teal, rose — and from v9.0 also a custom color chosen from a pop-up color palette), light, dark or automatic mode, and the switch for motivation features. The second group is the blood glucose thresholds: low limit, high limit and target value. The third group — new in v9.0 — is the time-of-day boundaries: here it can be set until what hour Breakfast, Mid-morning snack, Lunch, Afternoon snack and Dinner last (this drives the automatic meal-type suggestion), and until when the bolus calculator uses the morning and the midday ICR. The fourth group is insulin: the names of the rapid (bolus) and basal insulin — selectable from v9.0 in a drop-down list of the products available in Hungary, with a custom name still possible —, the action time in hours, the sensitivity, and the carb-to-insulin ratio (ICR) by time of day. The table in chapter 6.6 assists in completing the values together with the treating physician. From v10.0 a custom background color can also be chosen from a pop-up color palette: the selected color appears as a soft background gradient throughout the application, in both light and dark mode, while card contrast is preserved; it can be cleared at any time with the Reset button.

4.8 New entry

Reached with the plus button in the header. The date and time default to the current moment but can be changed freely, so a forgotten morning reading can still be logged in the afternoon with its real time. The type can be Meal, Check (measurement only), Basal (Lantus) or Other activity (e.g. a walk). For meals the application pre-selects the meal type by time of day — based on the boundaries configured in Settings, for example Lunch at noon. Foods are added from the quick picker with one tap and the carbs add up. When a blood glucose value is entered, the application calculates an insulin suggestion (details in chapter 5), which can be accepted with the Apply suggestion button, or a different dose can be entered manually.

Typo protection (v9.0): if the entered blood glucose value falls outside the plausible range — below 2.0 mmol/l or above 25.0 mmol/l — the application asks before saving whether the value is really correct. This way an accidental 65 instead of 6.5 does not distort the statistics and the report. Answering Yes still saves the unusual value, because rare but real extreme readings do occur.

4.9 Quick reference: buttons, icons, indicators

The table summarises the recurring elements of the screens to support confident use even in the first days. In v9.0 the icons follow a consistent drawn style.

Element	Where it is found	What it does
Plus button (New)	Header, top right	Records a new entry from any tab.
Bottom navigation bar	On phones, at the bottom of the screen	Switches between the seven tabs; the active tab is coloured.
Pencil icon	On entry cards	Edits the entry afterwards, including its time.
Bin icon	On entry cards	Deletes, after a confirmation question.
Green, amber, red colours	On cards and charts	In-range, low, and high blood glucose values.
Pulsing IOB label	Overview, insulin tile	Insulin is currently active in the body; the amount is shown in units.
Purple bar under the header	In follower mode	The follower’s view: the time of the last sync and the Sync now button.
Apply suggestion button	New entry, purple box	Moves the calculated insulin dose into the field, after confirmation.
Create report button	Bottom of the Statistics tab	Printable medical report for the given from–to period.
Sync now button	Sync tab and follower bar	Immediate manual upload or refresh through Drive.

5. A typical day with the application — a practical example

This chapter walks through an ordinary weekday from morning to evening through the eyes of an example user, to show where the diary fits into everyday life. The numeric values are illustrations; personal parameters must always be agreed with the treating physician.

7:00 am — waking measurement and breakfast

The user measures their blood glucose: 6.2 mmol/l. They press the plus button. The type is Meal and the meal type already says Breakfast. They enter 6.2, then pick from the quick picker: 2 slices of wholemeal toast (24 g carbs) and a medium apple (15 g carbs), 39 grams in total. The application calculates with the morning ICR (10 g/U in this example): 3.9 units; the deviation from target is small so the correction is negligible, there is no active insulin, and the suggestion is 4.0 units of rapid insulin. The user accepts the suggestion, injects the insulin, then Save. The Overview shows: 1/4 entries today, the streak continues.

12:30 pm — lunch with higher glucose

Reading: 9.0 mmol/l; lunch: chicken with rice, about 60 grams of carbs. The calculation has three parts. Carb part: 60 / 10 = 6.0 units. Correction: (9.0 measured minus 6.0 target) / 2.5 sensitivity = 1.2 units. Active insulin: nothing remains from the morning dose, so 0. The suggestion: 6.0 + 1.2 = 7.2, rounded to half units: 7.0 units. The application also displays the details, so the follower can see afterwards exactly how the dose was composed.

4:00 pm — treating hypoglycaemia

The user feels weak and measures: 3.4 mmol/l. They act first: 15 grams of glucose, 15 minutes of waiting, re-check: 4.6 mmol/l. Then they record the 3.4 value with the Check type. On saving, the application confirms the protocol in a pop-up (15 grams of fast carbs, re-check in 15 minutes), the value appears as an amber dot on the chart, and the follower’s phone receives an alert if it has been enabled.

9:30 pm — basal insulin and closing the day

The Overview signals in a purple bar: no basal insulin logged today. The user injects their usual basal dose and records it as a Basal-type entry. The daily circle turns 4/4, the weekly TIR tile shows 81%, and the day ends with a green confirmation.

Weekend example: exercise and a restaurant

On Saturday morning the user cycles for an hour and a half. Before setting off they measure 7.8 mmol/l, and because exercise lowers blood glucose, they eat 20 grams of carbohydrate without insulin according to the protocol agreed with the physician. They record this as an Other activity entry with the note: before 90 min cycling. In the evening they eat out. The exact carb content is an estimate in such cases: the user takes 70 grams as the basis of the bolus for the pasta dish, based on experience. After saving they write in the note: restaurant estimate, carbonara. Two hours later they log a check reading, 9.6 mmol/l. When the application calculates the next suggestion, it automatically subtracts the insulin still on board (IOB), reducing the danger of stacked doses. Days annotated like this are especially valuable at the clinic: the treating physician can see exactly which situations cause the swings.

Before the quarterly check-up: a report in two taps

On the evening before the appointment, the user opens the Medical report card on the Statistics tab, sets the past 90 days, presses Create report and saves the summary as a PDF. The next day the physician sees the TIR, the estimated HbA1c and GMI, the variability measures, the daily pattern and the full log on a single page — so the consultation can focus on decisions.

6. The phases of installation

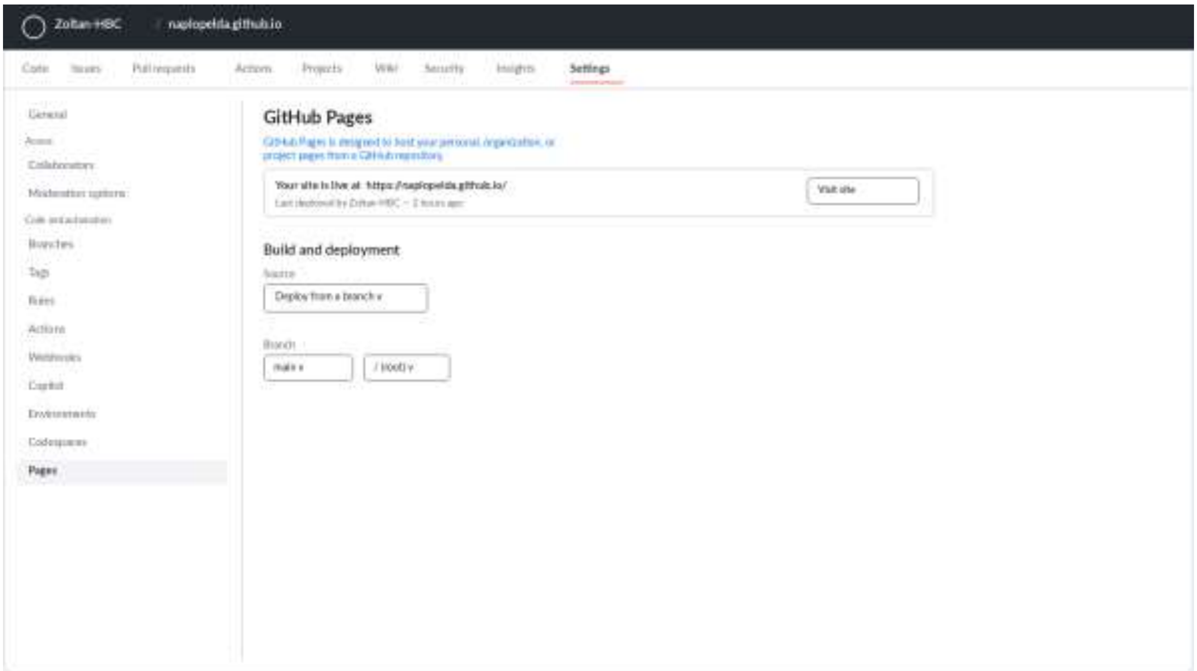
Installation has two big steps: first the application must be published once on the internet (it is sufficient for a single person to do this), then everyone installs it on their own device from a browser. The reason for publishing: for security reasons, installability and Drive sync only work on an encrypted, so-called HTTPS address — not when the file is opened directly from the computer.

6.1 First phase: publishing on GitHub Pages

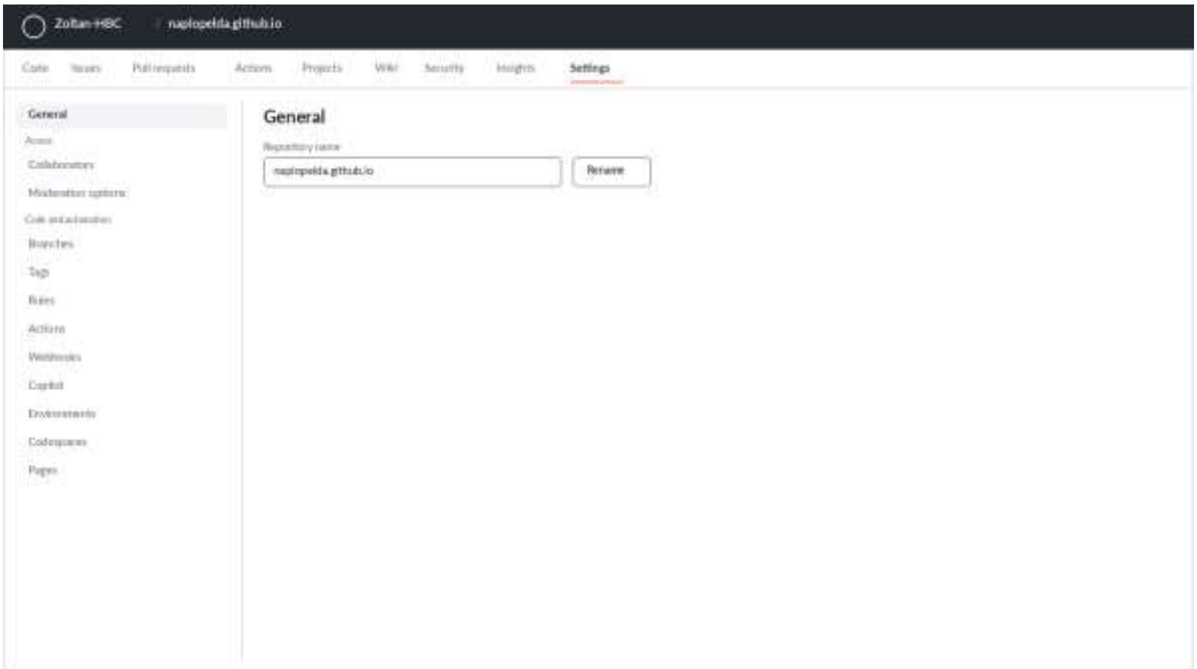
GitHub is the world's largest free code host, and its Pages service provides a free web address. No bank card is needed and the whole process takes about 15 minutes. Based on practical experience, the steps below highlight the two most common pitfalls: the exact repository name and the correct way of uploading the files.

- Open github.com and sign up (an e-mail address and a password are required).
- In the top right corner click the plus sign, then New repository.
- **The repository name must be exactly the full web address: YOURUSERNAME.github.io — including the .github.io ending! As an example, using a fictional username naplopelda: correctly, the repository name is naplopelda.github.io, and the application will be available at the full address https://naplopelda.github.io. Incorrect examples: just naplopelda (without the .github.io ending), or any other name that does not exactly match the username (e.g. naplopelda-site) — with these the Pages service will not start at the main address.**
- Set it to Public, then click Create repository.
- **Verification: in the repository's Settings → General section the name (in the example above: naplopelda.github.io), and on the Settings → Pages screen the web address, must both appear in the full https://naplopelda.github.io form — together with the .github.io part. If the ending is missing in either place, the name given in step 2 is wrong and the repository must be renamed (Settings → General → Repository name).**
- Click the "uploading an existing file" link.
- **When uploading, drag in ALL files and subfolders that are INSIDE the HBC_App_v11 folder — but NOT the folder itself! Correctly: open the HBC_App_v11 folder on your computer, select everything in it (Ctrl+A) and drag this selection into the browser window: the index.html, manifest.json and sw.js files and the css, js, lib, fonts, icons and store folders. Incorrectly: if you drag in the HBC_App_v11 folder as a whole, the application ends up under the …/HBC_App_v11/ sub-address and will not start at the main https://YOURUSERNAME.github.io address.**

k) GitHub — Settings → Pages: the "Your site is live at…" confirmation



l) GitHub — Settings → General: the exact repository name can be checked here



- Click Commit changes and wait 2–3 minutes.
- Open the full address https://YOURUSERNAME.github.io in a browser. If the application loads, publishing is complete.

Good to know

The public repository contains only the application's program files, not the diary data. The data is always stored on the individual phones and in the user's own Google Drive.

6.2 Installing on an Android phone

- Open https://YOURNAME.github.io in the Chrome browser.
- An Install offer pops up at the bottom of the screen. If not, tap the three dots in the top right, then Install app.
- Tap Install. The icon (the blue-grey ribbon logo) is placed on the home screen.
- Open it from the icon: the application runs full-screen and from then on works without internet as well.

6.3 Installing on an iPhone

- Open https://YOURNAME.github.io in the Safari browser.
- Tap the Share button at the bottom (a square with an upward arrow).
- Scroll down, choose Add to Home Screen, then tap Add.
- The icon appears on the home screen; started from there it runs full-screen, even without internet.

6.4 Installing on a Windows or Mac computer

- Open https://YOURNAME.github.io in the Chrome or Edge browser.
- A small install icon appears at the right edge of the address bar (a monitor with a downward arrow). Click it.
- Click Install. The application runs in its own window and is added to the Start menu or the Dock.

6.5 Taking over the data of an earlier (v6, v7 or v8) diary

There are two cases. If the new application is opened in the same browser in which the v6, v7 or v8 diary was used so far, the data appears by itself — nothing needs to be done. When starting on another device or browser: in the old diary open the Sync tab, press JSON Export, send the file to yourself (for example by e-mail), then on the Sync tab of the new application load it with the JSON/CSV Import button. The import is intelligent: it does not duplicate entries that already exist.

6.6 First settings, together with the treating physician

After installation open the Settings tab and complete the table below. The values in the left column are approved by the treating physician, because the insulin suggestions are calculated from them.

Setting	What it means	Example value (to agree with the physician)
Low blood glucose limit	The hypo warning sounds below this	3.9 mmol/l
High blood glucose limit	Values above this count as high	10.0 mmol/l
Target glucose	The target of the correction calculation	6.0 mmol/l
Sensitivity	The glucose-lowering effect of 1 unit of insulin	2.5 mmol/l
ICR morning	How many grams of carbs 1 unit covers in the morning	10 g
ICR midday	How many grams of carbs 1 unit covers at midday	12 g
ICR evening	How many grams of carbs 1 unit covers in the evening	10 g
Time-of-day boundaries	The hour limits of meal types and ICR periods	Breakfast until 9, Lunch until 14, Dinner until 21; ICR: morning until 10, midday until 16
Insulin action time	How long injected rapid insulin counts as active	4 hours
Insulin names	The names of the preparations used	e.g. Humalog and Lantus
Custom background color (v10.0)	A background color choosable from a pop-up palette next to Settings → Appearance	e.g. #EEF2FF (clear anytime with Reset)

6.7 Updates in the future

When a new version of the application is ready, updating is simple: the changed files are uploaded to the GitHub repository the same way as in chapter 6.1. The copies installed on phones and computers notice the new version by themselves: they download it the next time they are opened with an internet connection, and the version after that restart is already the new one. Diary data remains untouched during the update, because it lives not in the program files but in the device's own storage. Example: files uploaded on Tuesday evening are downloaded by the follower's phone when opened on Wednesday morning, and the second opening that morning already runs the new version.

7. Google Drive sync: the follower sees the values live

7.1 How it works

Sync is built on two roles. The owner's application automatically uploads the diary as a single file to a chosen folder of their own Google Drive after every save. The follower's — a relative's or the treating physician's — application connects to this shared file: it downloads fresh data on opening and at a configurable interval (5 minutes by default), and shows it on the same screens as the owner's — just read-only. Alerts can be requested too: when a fresh low or high value arrives, the follower's phone shows a notification. Alerts arrive while the application is open or in the background; with the application fully closed they do not — that would require a central server, which may be a future extension.

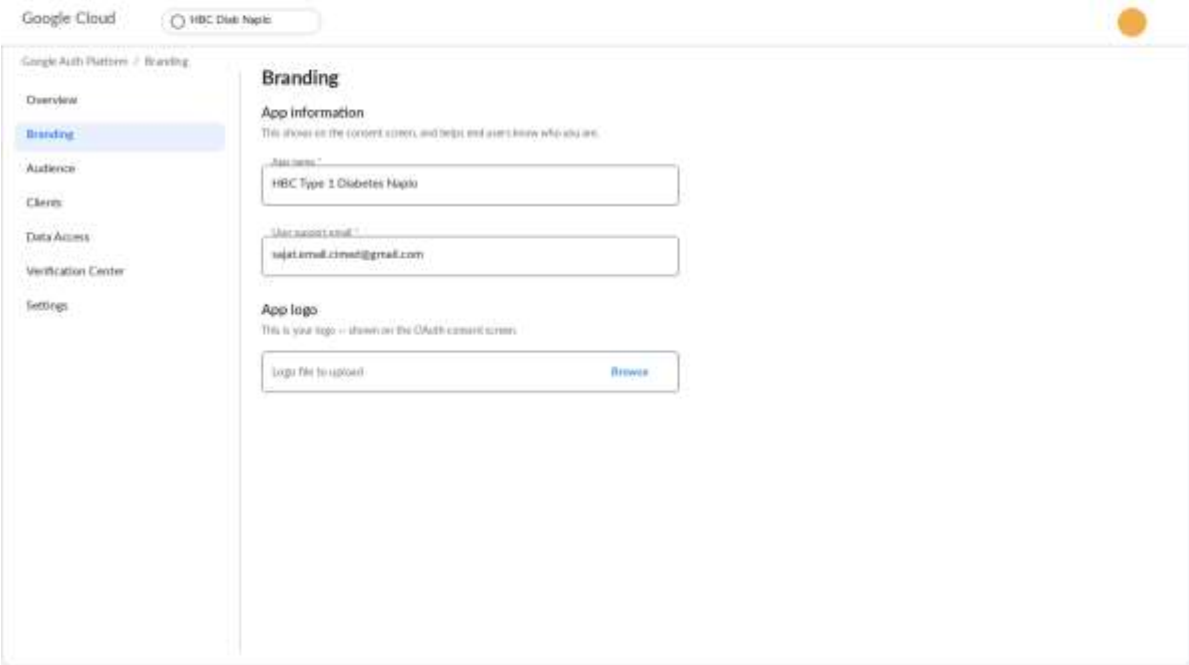
The application uses Google's strictly limited permission called drive.file: it can access only its own diary file and cannot even see the other files stored on the Drive.

7.2 One-time Google setup (about 20 minutes, free)

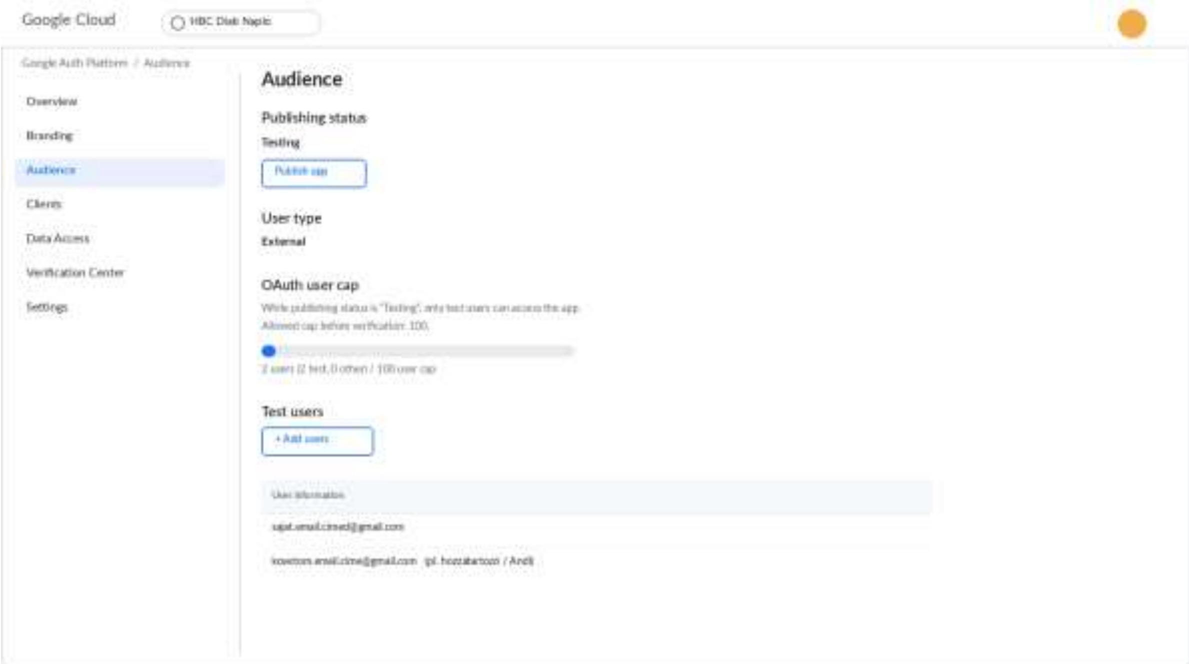
Google requires every such application to have a so-called Client ID. It must be created once, without a bank card.

- Open console.cloud.google.com and sign in with your Google account.
- At the top, in the project selector, click New project, name it HBC Naplo, then Create.
- In the left menu: APIs & Services, then Library. Find and enable (Enable) the Google Drive API, then the Google Picker API the same way.
- On first entry the Google Auth Platform setup wizard appears (Overview tab, Get started button). Enter the app name (e.g. HBC Diabetes Diary) and your email address (User support email); on the next step choose the External audience type, enter your email again as the contact, and accept the terms (Finish, Continue).
- In the left-hand Clients menu click + Create client. Set the type to Web application; any name is fine (e.g. Web client 1). In the Authorized JavaScript origins field enter your own address: https://YOURNAME.github.io, then Create.
- Click the created client to reveal the Client ID (a long string ending in .apps.googleusercontent.com) — copy it, it will be needed in step 7.3.
- In the left-hand Audience menu scroll to Test users and use + Add users to add your own email address and the follower's email address. In Testing publishing status up to 100 test users can be added, no verification required.

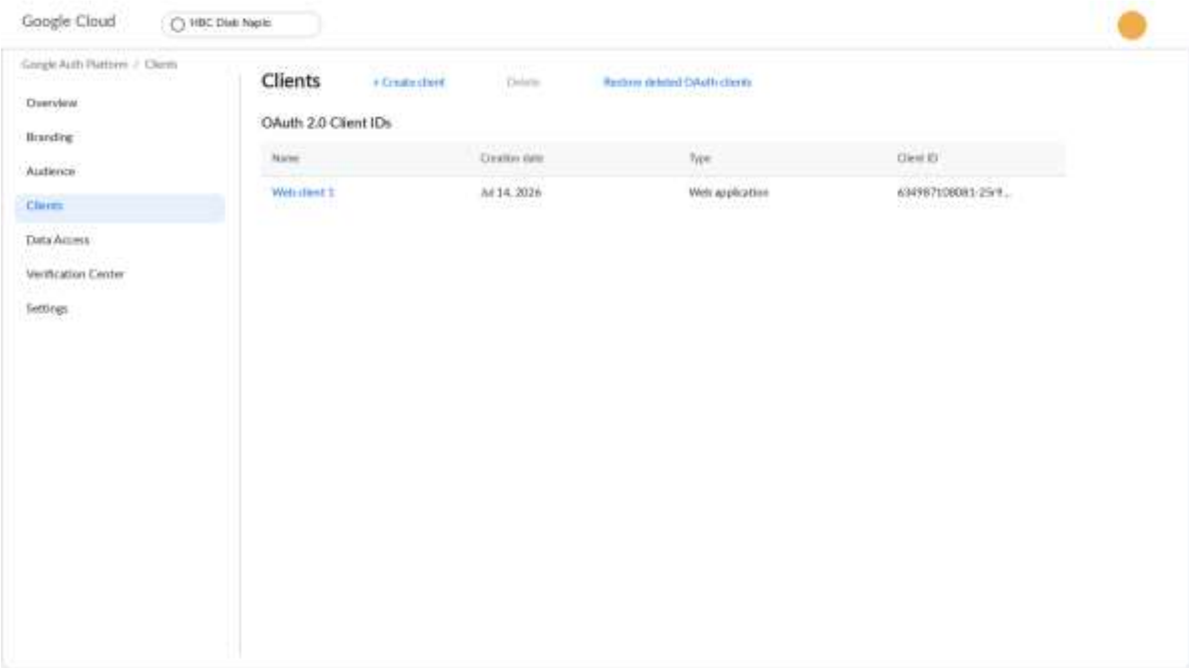
g) Google Auth Platform — Branding (app name, contact email)



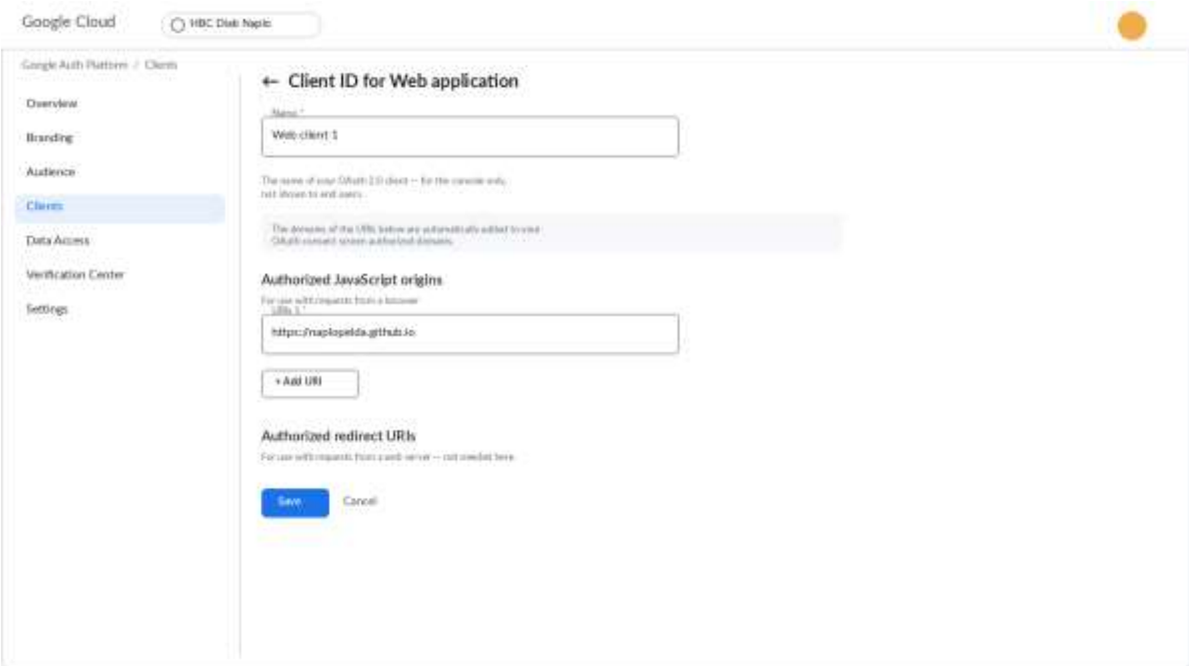
h) Google Auth Platform — Audience (Testing status, test users)



i) Google Auth Platform — Clients (the created OAuth client)



j) Client details — Authorized JavaScript origins and the Client ID



7.3 Setting up the owner (owner mode)

- In the HBC Diabetes Diary application itself — NOT in Google Cloud, in the diary app — open the Sync tab, and on the Google Drive sync card choose the Owner mode button.
- Paste the Client ID into field 1.
- Press button 2, Connect with Google account, and allow access. If the Google Cloud project is still in Testing status, Google may show an "unverified app" screen: proceed with Continue, because only the registered test users have access. In production status this screen does not appear.
- Press button 3, Choose folder, and select the diary folder on the Drive (worth creating in advance, for example named HBC Naplo).
- Done: from then on the file on Drive refreshes a few seconds after every save. It can also be triggered manually with Sync now.

7.4 Sharing the folder and setting up the follower (follower mode)

- On drive.google.com right-click the diary folder, choose Share, enter the follower's e-mail address with the Viewer role, and send it.
- The follower installs the application on their own phone the same way (chapter 6.2 or 6.3).
- On the Sync tab the follower chooses the Follower mode button, pastes the same Client ID, and connects with their own Google account.
- Tapping the Choose shared diary button, they select the HBC_Diabetesz_Naplo.json file from the Shared with me section of the picker.
- In the Alert settings box they enable notifications, switch on the low-value alert, and set the refresh interval (for example 5 minutes).
- Done: the purple bar under the header shows the time of the last sync, and the Sync now button requests an immediate refresh.

An everyday example

The owner records a 3.4 mmol/l value at 16:05. The diary is uploaded to Drive at 16:05. The follower's phone downloads it at the next refresh, by 16:10 at the latest, and since the value is below the 3.9 limit, it shows an alert: HBC Diary, low value alert, 3.4 mmol/l.

Important (v9.0): the alert fires only for the latest measurement taken on the current day. A back-dated value entered for an earlier day never triggers an alert, so there is no false night-time wake-up because of a past record being filled in. The shared data package also includes the CGM sensor readings, so the alert always reflects the real, current state.

7.5 Backup in practice: the monthly routine

Drive sync gives continuous protection, but it is worth adding a simple monthly routine, because a manual backup is independent even of the Google account. The suggested procedure on the first Sunday of every month: on the Sync tab press JSON Export; the downloaded file name automatically contains the date (for example HBC_v9_backup_2026-08-02.json); copy it to a USB stick or send it to yourself by e-mail. If everything ever needs restoring (new phone, lost device, cleaned browser), the latest such file can be loaded back within minutes with the JSON/CSV Import button, and the import does not duplicate existing entries. The Overview screen warns when the latest backup or sync is older than 7 days, so skipping the routine does not go unnoticed.

8. Loading CGM data (for sensor users)

When using a continuous glucose sensor (for example FreeStyle Libre or Dexcom), the measurement file downloadable from the manufacturer's system can be loaded into the application and appears as a separate curve on the Charts tab. There is currently no live, direct connection to the sensor, because the manufacturers do not open this to third-party applications, but the application is prepared for it, so a later extension can add it.

LibreView example

- Sign in at libreview.com and save the CSV file with the Download glucose data button.
- On the application's Sync tab press the Choose CGM CSV file button and select the downloaded file.
- The confirmation shows the number of imported readings, for example: CGM: 1340 readings imported.
- The CGM sensor curve appears on the Charts tab, with the target-range band.

Dexcom Clarity example

At clarity.dexcom.eu a CSV file can be requested with the Export button; loading works the same way. The application recognises the file's unit: values saved in mg/dl are converted to mmol/l automatically, so the data is stored consistently.

9. Publishing in the Google Play store

The application can also be uploaded to the Play store, so anyone can install it from the store by searching. This is not a requirement for use (installation as in chapter 6 is fully featured); it is rather the path of an official release for a wider audience. The materials needed for the process are ready in the HBC_App_v11/store folder: store texts in Hungarian and English, a privacy statement, and a packaging configuration file.

9.1 Creating a developer account

- Open play.google.com/console/signup and sign in with a Google account.
- Choose the Personal account type and pay the one-time 25 dollar registration fee by bank card.
- Complete the identity verification (a photo of an ID card). Approval usually takes a few days.

9.2 Building the Android package

A packaged Android file (AAB) is uploaded to the Play store; it is produced from the published application by the free tool Bubblewrap. This requires Node.js on the computer (nodejs.org, LTS version).

- In a command prompt run: npm install -g @bubblewrap/cli
- Then: bubblewrap init --manifest=https://YOURNAME.github.io/manifest.json. For the questions, the values in the store folder's twa-manifest.json are authoritative.
- The process creates a signing key: write down and keep its passwords, they will be needed for every future update.
- Run: bubblewrap build. The result is the app-release-bundle.aab file.
- Bubblewrap prints content named assetlinks.json: upload it to the website into the .well-known folder. Without it a browser bar would show inside the app.

9.3 Upload and store listing

- In the Play Console click Create app; the name is HBC Diabétesz Napló, the language Hungarian.
- In the Internal testing track create a release and upload the AAB file. This gives an immediately installable link for trying it out before the public release.
- Copy the texts of store/STORE_ANYAGOK.md to the store listing (short and long description in Hungarian and English), upload the 512-pixel icon and the screenshots.
- On the data safety form declare: the application does not collect and does not share data with third parties. Provide the web address of the privacy statement.
- In the health declaration state that the application is not a medical device.
- After successful testing create the production release and submit it for review. Assessment typically takes 1–7 days.

Apple App Store note: for iPhones, installation as in chapter 6.3 is the recommended route. An App Store release requires an Apple developer account (99 dollars per year) and a Mac computer, so it warrants a separate decision.

10. Troubleshooting: frequent questions

Symptom	Cause and solution
The Install button does not appear on the phone.	On Android only Chrome offers it, on iPhone only Safari. Check that the address is https://YOURNAME.github.io and the file was not opened directly.
The application does not update to the new version.	Close the application completely, then open it again with an internet connection. The update activates on the second opening.
The medical report does not open.	The report opens on a new page: allow pop-ups for this address in the browser settings, then press Create report again.
The application questions the value when saving.	This is the typo protection: it asks for confirmation before values below 2.0 mmol/l or above 25.0 mmol/l. A real extreme value can be saved with Yes.
An “unverified app” screen appears at the Drive connection.	This only appears if the Google Cloud project is still in Testing status — in that case proceed with Continue, but only registered test users have access. In production status (see chapter 7.2) this screen does not appear.
The follower cannot find the shared file in the picker.	Check that the folder on Drive is shared with the follower’s exact e-mail address and that the owner’s application has synced at least once (that is when the file is created).
No alert arrives on the follower’s phone.	In the Alert settings the Enable notifications button must be green, and the application must be open or running in the background. With the application fully closed the alert does not arrive.
Drive asks to sign in again every hour.	By Google’s security rule, access renews roughly every hour. This usually happens by itself, without a pop-up window.
The CGM file import reports an error.	Check that the original, unmodified CSV file from LibreView or Clarity is being loaded. A file opened and re-saved in Excel may have a changed structure.
Data disappeared after cleaning the browser.	The application’s dual storage (backup database) usually restores it automatically on restart. If not, everything can be reloaded from the latest JSON backup with the Import button.
A few Hungarian words remain in English mode.	Manually entered data (food names, notes, insulin names) intentionally stays exactly as recorded — that is the user’s personal data.
The Sync now button shows an error message (for example “Login expired”, “No permission”, “File not found”, “No internet connection”).	This precise error message (v11.1) helps pinpoint the problem: for an expired login, sign in again with Connect with Google account; for a missing permission, check that the folder is shared; for a deleted file, choose the folder again; without an internet connection, the data syncs on the next successful attempt. The scheduled, automatic sync retries silently in this case, without showing an error.

11. Medical and privacy notes

11.1 From a specialist's point of view

The application's dose suggestion follows the internationally common three-part formula: the carbohydrate coverage (total carbs divided by the ICR) plus the correction (the difference of the measured and target values divided by the sensitivity) minus the insulin on board (IOB). The ICR can be set separately by time of day — and in v9.0 the hour boundaries of these periods can be adapted to the patient's routine —, the action-curve length is adjustable, and the result is rounded to half units. Every parameter is to be entered on the Settings tab under medical supervision. The estimated HbA1c is computed from the average glucose of the selected period with the common conversion formula; alongside it, v9.0 also calculates the GMI (Glucose Management Indicator) with the international formula. The application consistently marks both as estimates. The medical report provides a summary of any period that can be printed or forwarded as a PDF, with SD/CV variability measures. Follower mode is read-only: the treating physician sees the patient's data but cannot modify it.

What to review during the appointment: a checklist

The following list gives a fast, factual overview from the application at the quarterly check-up, illustrated with examples.

- Medical report for a 90-day period: every key figure and the detailed log on a single printable page — the fastest starting point.
- Statistics tab, 90-day range: TIR, average, estimated HbA1c and GMI together. Example: with 76% TIR and 7.1% estimated HbA1c the direction of therapy is good; details can be refined on the charts.
- Daily pattern chart: the hourly averages. Example: if the 3–6 am band is repeatedly low, lowering the basal dose may come up — a decision for the specialist.
- Measurement summary: the ratio of low, normal and high readings. Example: 3 low, 41 normal, 9 high indicates that the excursions tend upward.
- Carb/insulin ratio card: the actual ratio computed from the diary against the configured ICR. Example: if the configured morning ICR is 10 but the diary shows 1 unit covering only 8 grams, adjusting the ratio may be justified.
- Entries tab with the notes: the context of restaurant, sport and stressful days, as in the examples of chapter 5.

11.2 Where the data is

Every entry is stored on the given device, with dual storage: besides the fast primary store, a backup database also keeps it, so it recovers even after an accidental browser clean-up. The application sends no data whatsoever to the developer or any third party, and contains no advertising or tracking code. With Drive sync enabled, a copy of the diary also exists on the user's own Google Drive, and only those with whom the user themselves shared the folder can access it. Sharing can be revoked at any time on the Drive interface, and sync can be switched off with the Disconnect button. The medical report is generated locally, on the device — its content reaches anyone only if the user prints or sends it themselves.

11.3 Summary

The HBC Diabetes Diary v11.1 Type 1 Diabetes APP is an installable, offline-capable, bilingual diary that turns logging into calculation, feedback and sharing: it suggests insulin doses based on values agreed with the physician, warns at low and unusual values, shows the trends on charts, statistics and the daily pattern, produces a printable medical report for any period, and can be followed live on the designated follower’s phone. Installation consists of a one-time publication and a few taps per device; Drive sync works by itself after a one-time Google setup. The chapters of this document walk through every phase from everyday use to the store release, and chapter 10 resolves the most frequent difficulties.